Questions from May 31, 2025

Question 1

In proposition 2.9 in my thesis i simply cannot understand how that *Chain of points* construction works. I have tried to prove it myself without it. What i found was: Let $x, y \in V^*$ and ϵ given. Then $x \in V_p$ and $y \in V_q$ then WLOG assume $p \leq q$, thus $x \in V_q$. We can therefore construct a path of neighbours from x to y so

$$x \underset{q}{\sim} \dots \underset{q}{\sim} y$$

Now we are interested in the length of this. Now we know if $x \underset{m}{\sim} y$ then $|x - y| \leq 1/2r^m \cdot \mathrm{diam}(K)$, so the total number of jumps need are

$$jumps = \frac{\delta}{1/2r^q \cdot \mathrm{diam}(K)} \leq \delta \cdot 2 \cdot r^{-q}$$

(here δ is part of the ϵ-δ proof), and *jumps* is only correct if it is a straight line, but as the amount of jumps are finite we can multiply by a constant C_1 and find thus by triangle inequality and many $\pm$ we get

$$\left|u(x) - u(y)\right| \leq jumps \cdot r^{q/2} \mathcal{E}^{1/2} \leq C_1 \delta \cdot 2 \cdot r^{-q} r^{q/2}$$

and from here i simply don't know how to uniformly bound it.

Question 2

In lemma 2.35 in my thesis, i am stuck at showing that

$$\int_K G_s(x, y)\ \mathrm{d}\mu(y) < \infty$$

when $s = d_h/d_w$. I understand the proof when $s > d_h/d_w$ as this is a contant and $s < d_h/d_w$ since by constructing the open balls $B(x, R2^{-j})$ we can bound $s^{-j-1} \leq R^{-1} d(x, y) \leq 2^{-j}$ with $R = \mathrm{diam}\, K$ and follows from there. But i simply cannot find a way to bound $\left|\ln(d(x, y))\right|$.

Question 3

In lemma 3.6 of my thesis. I still don't understand how they get that last equality. I keep trying to do some decomposition into the basis but i come up short v_k and v_j a subtracted and not multiplied together.

Question 4

Why does the discrete laplacian Δ_m generate a semi-group. I tried to use theorem 2.23 from my thesis, but wondered if that is overkill and there is a simpler explenation.